# Dylan McDermott

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## EDUCATION

Iowa State University, Ames, IA — January 2022 - May 2024
*Master of Science, Meteorology* — GPA: 3.52/4.0
- Thesis: Impacts of U.S. Deforestation on Rainfall from Mesoscale Convective Systems

Iowa State University, Ames, IA — August 2018 - May 2022
*Bachelor of Science, Meteorology*

## TECHNICAL SKILLS

- Numerical Modeling & Simulation: WRF and Noah-MP execution, high-resolution atmospheric and land-surface modeling
- Geospatial Engineering: Spatial ETL, automated pipelines, raster/vector processing, and web GIS
- Meteorological Analysis: Severe weather verification, catastrophe modeling, and general meteorology
- Languages & Platforms: Python, SQL, R, MATLAB, C++, ArcGIS Pro/Enterprise, AWS, Linux, WRF
- Data Operations & Systems: Automated QC/validation, REST API integrations, and continuous operational workflows
- Environmental Sensing: Field instrumentation, GPS-linked mobile sensors, and automated systems

## WORK EXPERIENCE

***Meteorologist*** — January 2025-Present
Property & Liability Resource Bureau, Des Moines, IA

- Build and maintain production Python and ArcPy pipelines that ingest and validate NOAA, IEM, radar, and model data — automating hours of manual daily work while keeping maps, exports, and ArcGIS server updates synchronized.
- Automate the processing of hundreds to thousands of reports, files, and features per day while reconciling late reports and detecting duplicates, missing timestamps, invalid identifiers, incomplete location fields, and silent download failures.
- Develop claims-facing ArcGIS Experience Builder applications for hail, wind, hurricanes, and current weather, including custom REST API integrations for date filtering, layer control, map synchronization, and location-specific reports.
- Evaluate new datasets for scientific defensibility, spatial and temporal resolution, bias, provenance, update behavior, and appropriate interpretation; distinguish observations from modeled and remotely sensed estimates.
- Contributed to PLRB receiving the 2025 Esri Special Achievement in GIS (SAG) Award for innovative catastrophe data automation and web GIS applications.

***Field Sensing Research Associate (Contract)*** — June 2024-January 2025
Corteva Agriscience, Johnston, IA

- Co-developed and operated the Smartstick, a wheeled field-sensing platform that collected thousands of timestamped, Trimble GPS-linked canopy and air-temperature measurements per walk across seven research sites.
- Independently built the Python and ArcPy pipeline from raw tables through visual quality control, outlier and GPS filtering, point creation, inward plot buffers, spatial joins, maps, tables, shapefiles, and plot-level summaries.
- Automated approximately one hour of processing per collection, ultimately weeks of processing across the campaign, and applied one reproducible workflow across sites.
- Compared field measurements with drone imagery, LiDAR-derived canopy structure, treatments, and crop-stress response to validate spatial patterns and research findings.
- Co-designed and built a 16-chamber automated soil-gas system, a multi-instrument reference weather station, and tracking dashboards for more than 200 deployed infrared radiometers.

***Graduate Research Assistant*** May 2022-May 2024
Iowa State University, Ames, IA

- Designed and ran WRF and Noah-MP experiments at multiple resolutions to quantify how U.S. land-use change affected Midwest rainfall, surface fluxes, moisture transport, and mesoscale convective systems.
- Prepared model inputs by translating CESM/LUMIP land-use data to WRF and Noah-MP grids and evaluating present-day versus 1850 vegetation scenarios.
- Processed and compared ERA5, CESM, WRF, GOES, MODIS, MRMS, radar, and observational datasets using Python, MATLAB, and NetCDF files.
- Produced geospatial analyses, derived fields, maps, scientific visualizations, technical writing, and presentations for interdisciplinary research.

***Undergraduate Research Assistant*** May 2021-August 2021
Iowa State University, Ames, IA

- Conducted boundary-layer meteorology research by identifying cases, handling large observational and gridded atmospheric datasets, and producing visualizations to interpret physical mechanisms.

## PERSONAL PROJECTS

AI Weather Intelligence Platform 2025-Present

- Designed a full-stack proof-of-concept as a self-driven initiative to bridge atmospheric data systems with applied AI, building end-to-end prototypes that earned semi-finalist recognition in the AWS 10,000 AIdeas competition.
- Built working prototypes demonstrating automated evidence extraction and report generation.